SMART INDIA HACKATHON 2026 · TECHNICAL REPORT

KADAL NETRA

கடல் நேத்ரா — the eye of the sea

An Indian-domain side-scan sonar intelligence pipeline for detecting, verifying and geolocating anthropogenic marine debris — producing ranked, dive-ready cleanup targets instead of bounding boxes.

DOMAIN	THEATRE	BUILD WINDOW	PREPARED
Marine Perception	**Gulf of Mannar**	**36 hours**	**30 Aug 2026**
Side-scan sonar AI	Palk Strait sector	Grand finale prototype	Revision 1.0

00 Contents

01 Product Identity and Naming

The product is **Kadal Netra** — கடல் (kadal, "sea") + नेत्र / நேத்ரா (netra, "eye"). It reads cleanly in Tamil and Hindi, needs no explanation on a slide, and carries the exact functional promise of the system: sight where there is none.

CANDIDATE	READING	ASSESSMENT
Kadal Netra CHOSEN	"Eye of the sea"	Tamil-rooted, pan-Indian legibility, matches the Gulf of Mannar theatre. Short, pronounceable by a non-Tamil jury. No trademark collision found in marine-tech.
Varuna-SSS	Vedic sea deity	Institutional register; aligns with existing Indian defence naming conventions. Slightly generic — "Varuna" is heavily reused.
Neera	நீர் — water	Very short and brandable, but reads as a consumer app, not a survey instrument.

CANDIDATE	READING	ASSESSMENT
GhostGrid	English-descriptive	Immediately explains the georeferenced output grid. Loses the Indian-domain positioning that is the project's core defensible claim.

Naming the subsystems

Give the internal modules names too. It makes the architecture slide legible in three seconds and signals engineering maturity.

MODULE	FUNCTION	WHY THE NAME
Alai (அலை)	Sonar ingestion and preprocessing	"Wave" — the raw acoustic front door.
Nizhal (நிழல்)	Shadow-geometry physics verifier	"Shadow" — literally what the module reasons about.
Valai (வலை)	Ghost-net segmentation head	"Net" — the primary target class.
Thadam (தடம்)	Multi-ping track association	"Trace / track" — turns detections into physical objects.
Padai (படை)	Cleanup-priority planner and dive-plan export	"Deployment / force" — what gets sent to the water.

02 System Thesis — What We Are Actually Building

Almost every submission on a sonar problem statement will be YOLO fine-tuned on a public dataset with a Leaflet map bolted on. That is table stakes and it will not win. Kadal Netra wins on a different axis: it treats detection as an *evidence-accumulation problem under domain shift*, and it delivers an operational artefact — a ranked dive plan — rather than an accuracy number.

> THE ONE-SENTENCE PITCH
>
> Kadal Netra reduces the volume of sonar imagery a survey operator must manually inspect by an order of magnitude, while maintaining high recall on genuine anthropogenic debris in Indian coastal waters — and outputs a georeferenced, priority-ranked cleanup plan an NGO dive team can act on the same day.

Three commitments that shape every technical decision

1. The metric is false alarms per surveyed square kilometre, not mAP. An operator drowning in alerts will switch the system off. We optimise, report and demo against operator burden. Reframing the metric is itself part of the contribution.

2. Every detection carries auditable provenance. The confidence number is a composition of named evidence sources — detector score, shadow-geometry consistency, ping persistence, bathymetric plausibility. The UI shows the breakdown. A black-box score is unusable in a government survey workflow and indefensible under jury questioning.

3. Physics is a free prior and we use it. Acoustic shadow length is analytically related to object height, vehicle altitude and slant range. We enforce that relation as a hard filter. No overseas model in the prior art does this, it costs almost nothing to compute, and it is visually demonstrable.

What we explicitly do not claim

CLAIM DISCIPLINE

- We do **not** claim AI ghost-net detection is novel. GhostNetZero (Microsoft Research, DeepLabV3 + ResNet-50) demonstrated it publicly. We cite them as prior art in the deck — pre-empting the question is worth more than hoping it isn't asked.
- We do **not** claim sonar + navigation fusion is unattempted.
- We do **not** claim GEBCO provides object-scale bathymetry. At 15 arc-seconds it is regional context only.
- We do **not** mention Kumari Kandam. The archaeological framing invites a hostile question that cannot be won on stage. Palaeochannels appear in exactly one place: as a hard-negative class in the training taxonomy.

03 End-to-End Architecture

Modular, not monolithic. Each stage is independently testable, independently demonstrable, and independently degradable — if the segmentation head fails during the demo, the detector and the map still work.

0 · INGEST — Alai

`pyxtf · numpy · rasterio`

Raw XTF/JSF or pre-converted PNG + metadata CSV. Extracts per-ping: timestamp, slant range, frequency, heading, altitude, GNSS fix, port/starboard channel.

▼

1 · PREPROCESS — Alai

`opencv · scipy · numpy`

Slant-range correction, water-column removal, TVG / beam-pattern normalisation, CLAHE contrast equalisation, empirical-mode speckle suppression, overlapping 1024×1024 waterfall tiling with ping-index provenance preserved per tile.

▼

2 · DETECT

`YOLOv11-s (ONNX) · FiLM frequency conditioning`

Bounding-box candidate generation. Frequency and range-setting injected as a conditioning vector into early layers to attack domain shift directly.

▼

3 · SEGMENT — Valai

`SegFormer-B0 · fallback DeepLabV3+/ResNet-50`

Pixel masks for thin, irregular, tangled targets where a box is meaningless. Runs only on detector candidates, not the full frame.

▼

4 · PHYSICS VERIFY — Nizhal

`numpy · shapely — no neural network`

Shadow-geometry consistency check. Predicts expected shadow polygon from vehicle altitude and slant range; rejects candidates whose observed shadow implies an impossible object height.

▼

5 · ARTIFICIALITY VERIFY

`XGBoost over engineered features · ConvNeXt-Tiny optional`

Natural / Artificial / Unknown. Features: shadow ratio, edge linearity, Hu moments, GLCM texture, Gabor energy, fractal dimension. Feature importances are displayable — a defensible, non-black-box stage.

▼

6 · OPEN-SET SCREEN

`Energy-based OOD score · Mahalanobis on penultimate features`

Routes unrecognised artificial anomalies to human review rather than forcing them into a known class.

▼

7 · TRACK — Thadam

`scipy.optimize.linear_sum_assignment · PostGIS ST_DWithin`

Projects each detection to WGS-84, associates across pings and survey lines in geographic space, confirms a physical target at $N \geq 3$ supporting observations.

▼

8 · GEOSPATIAL CONTEXT

`rasterio · richdem · GEBCO_2026 GeoTIFF`

Samples depth, slope, curvature, terrain roughness at each target centroid. Used as a false-positive filter and as a priority input, never as primary detector input.

▼

9 · PRIORITISE — Padai

`explicit weighted scoring function`

Entanglement risk × reef proximity × diver accessibility × cluster density × calibrated confidence → ranked GeoJSON dive plan.

▼

10 · SERVE

`FastAPI · PostgreSQL/PostGIS · Next.js · MapLibre · deck.gl`

Waterfall viewer with mask overlay, map with clustered targets, evidence-provenance panel, confirm/reject/reclassify loop, CSV/JSON/GeoJSON/GPX export.

04 The Complete Technology Stack

Every layer, every library, with the reason it was chosen over the obvious alternative. Pin these versions — a dependency resolution failure at hour 30 of a 36-hour build is how teams lose.

4.1 Sonar ingestion and signal processing

COMPONENT	CHOICE	RATIONALE / NOTES
XTF parsing	`pyxtf 1.4+`	Only maintained pure-Python XTF reader. Handles Triton/EdgeTech headers. Post-hackathon priority — for the 36h build, ingest pre-converted PNG + metadata CSV.
JSF parsing	`pyjsf` / custom	EdgeTech native. Lower priority; only if the institutional data arrives in JSF.
Array ops	`numpy 2.1`	Baseline. Keep waterfall arrays as `float32` — `uint8` conversion destroys dynamic range needed for TVG correction.
Signal DSP	`scipy 1.14`	Savitzky-Golay for TVG curve fitting, `ndimage` for morphological shadow extraction, Hilbert transform for envelope detection.
Image ops	`opencv-python 4.10`	CLAHE, slant-range interpolation, connected components, contour extraction for shadow polygons.
Speckle suppression	`scikit-image 0.24`	Non-local means and BM3D-style denoise. Do not over-denoise — thin net strands are the first thing you destroy.
Tiling	Custom module	Write this yourself. 1024×1024 with 128px overlap, each tile carrying `(survey_id, ping_start, ping_end, channel, slant_range_m)`. This metadata linkage is where geolocation accuracy comes from and it is the single most commonly skipped piece.

4.2 Machine learning — training

COMPONENT	CHOICE	RATIONALE / NOTES
Framework	`PyTorch 2.5` + CUDA 12.4	Non-negotiable. Use `torch.compile` for a free ~20% training speedup on Ampere+.
Backbone zoo	`timm 1.0`	Access to ConvNeXt, EfficientNet, MobileViT with consistent APIs and pretrained weights.
Detector training	`ultralytics 8.3`	Fastest path to a working YOLOv11. CAUTION AGPL-3.0. Fine for a hackathon demo; flag the licence in the deck and note RT-DETR or a raw-PyTorch YOLOv8 reimplementation as the productionisation path.
Segmentation	`transformers 4.45` (SegFormer) + `segmentation-models-pytorch 0.3`	SMP gives DeepLabV3+, U-Net++, and FPN behind one interface for rapid ablation.
Augmentation	`albumentations 1.4`	Critical. Sonar-specific pipeline detailed in §6.6 — standard ImageNet augs are actively harmful here.

COMPONENT	CHOICE	RATIONALE / NOTES
Tabular verifier	`xgboost 2.1` + `scikit-learn 1.5`	Artificiality classifier over engineered features. Trains in seconds, ships feature importances, survives jury interrogation.
Calibration	`netcal 1.3`	Temperature scaling and reliability diagrams. Calibration is a claimed contribution — measure it properly.
Experiment tracking	`Weights & Biases` (free tier) or `mlflow`	W&B for the shareable run dashboard you can pull up mid-Q&A. Offline mode as fallback if venue wifi collapses.
HPO	`optuna 4.0`	Only for the XGBoost verifier — deep model HPO is not affordable in the window.
Data versioning	`DVC 3.x` or manifest JSON	For 36 hours, a checked-in `data/manifest.json` with SHA-256 per file is sufficient and has zero setup cost.

4.3 Synthetic data generation

COMPONENT	CHOICE	RATIONALE / NOTES
Net geometry	`trimesh 4.x` + custom drape solver	Generate mesh nets, apply gravity drape over synthetic seabed relief, randomise mesh size, tear patterns, fouling density.
Acoustic model	Custom NumPy — Lambert + specular	Backscatter `= A·cos²θ + B·specular(θ, roughness)`. Crude is fine. It only needs to be right about shadow geometry and intensity ordering.
Speckle statistics	`numpy.random` — Rayleigh	Sonar speckle is Rayleigh-distributed, not Gaussian. Getting this right is what makes synthetic frames transfer.
Compositing	`opencv` Poisson blend	Composite synthetic targets onto real Indian seabed backgrounds so the background statistics are genuine even when the object is not.
Optional high-fidelity	`Blender 4.2` headless + `bpy`	Ray-traced shadow casting if time permits. Massive visual credibility on a slide. Build before the hackathon, not during.

4.4 Inference and serving

COMPONENT	CHOICE	RATIONALE / NOTES
Runtime	`onnxruntime-gpu 1.19`	Export detector and segmenter to ONNX opset 17. Consistent CPU/GPU behaviour, no PyTorch in the serving container.
Edge accelerate	`TensorRT 10.x`	FP16 engine build. SKIP IN 36H unless a Jetson is physically present. Roadmap item.
Quantisation	`onnxruntime.quantization`	INT8 static quantisation with a sonar calibration set. Report the accuracy delta honestly.
Batching	FastAPI `BackgroundTasks`	Celery + Redis is the correct production answer and the wrong hackathon answer. BackgroundTasks with an in-process queue ships in 20 minutes.
Model registry	Local `models/` + SHA manifest	Every prediction record stores the model hash that produced it. Cheap reproducibility.

4.5 Geospatial

COMPONENT	CHOICE	RATIONALE / NOTES
Raster I/O	`rasterio 1.4`	GEBCO GeoTIFF sampling via `rasterio.sample`. Install via rasterio wheels only — never install standalone GDAL near a demo.
Vector	`geopandas 1.0` + `shapely 2.0`	Detection footprints, survey line geometry, MPA polygon intersection.
Projections	`pyproj 3.6`	WGS-84 ↔ UTM 43N/44N (Indian coastal zones). Do distance maths in UTM, store in WGS-84.
Terrain derivatives	`richdem` or 3×3 Sobel	Slope, aspect, curvature, TRI from GEBCO. A hand-rolled Sobel kernel avoids a fragile dependency.
Bathymetry source	GEBCO_2026 Grid + TID Grid	Download a user-defined Indian-water subset as GeoTIFF. Do **not** pull the 7.0 GB global NetCDF. The TID grid tells you the provenance quality of each depth cell — use it to weight bathymetric confidence.
Tiling for web	`rio-tiler` / pre-baked PMTiles	PMTiles is a single-file tile archive served statically. No tile server to babysit.

4.6 Backend

COMPONENT	CHOICE	RATIONALE / NOTES
Framework	`FastAPI 0.115` + `uvicorn`	Async, auto OpenAPI docs. The `/docs` page is a free "we have an API" demo moment.
Validation	`pydantic 2.9`	Strict schemas for detection records. Catches metadata corruption at the boundary.
ORM	`SQLAlchemy 2.0` async + `GeoAlchemy2`	GeoAlchemy2 gives you PostGIS geometry columns as first-class types.
Migrations	`alembic`	Set up in the first hour. Schema churn during a hackathon is guaranteed.
Driver	`asyncpg`	Fastest async Postgres driver.
File storage	Local volume, S3-compatible via `minio`	MinIO in Docker Compose if you want to demo the cloud story. Local volume otherwise.
Auth	`fastapi-users` or a static demo token	SKIP IN 36H A hardcoded operator token is fine. Nobody scores you on auth.
Streaming	Server-Sent Events	Push inference progress to the dashboard. Simpler than WebSockets and sufficient.

4.7 Database

COMPONENT	CHOICE	RATIONALE / NOTES
Primary	`PostgreSQL 16` + `PostGIS 3.4`	Spatial indexing, `ST_DWithin` for track association, `ST_ClusterDBSCAN` for debris hotspot clustering — clustering in SQL rather than Python is a genuine speed win.
Raster in DB	`postgis_raster`	Optional. Sampling GEBCO from a GeoTIFF on disk is simpler.

COMPONENT	CHOICE	RATIONALE / NOTES
Time-series	`TimescaleDB`	SKIP Ping-level telemetry does not need it at prototype scale.
Vector search	`pgvector 0.7`	DIFFERENTIATOR Store the penultimate embedding of every confirmed target. Enables "find me everything that looks like this" retrieval — a very strong 15-second demo moment.
Cache	`Redis 7`	Only if you add Celery. Otherwise omit.

4.8 Frontend

COMPONENT	CHOICE	RATIONALE / NOTES
Framework	`Next.js 15` App Router + `TypeScript 5.6`	Server components for the initial target list, client components for the map.
Map	`MapLibre GL JS 4.x`	Open-source, no Mapbox token, vector tiles, GPU rendering. Basemap: OpenFreeMap or a self-hosted PMTiles bathymetry style.
Map overlays	`deck.gl 9`	ScatterplotLayer for targets, HeatmapLayer for debris density, PathLayer for survey lines, PolygonLayer for MPA boundaries.
Waterfall viewer	`OpenSeadragon 4.x`	HIGH LEVERAGE Deep-zoom on a 40,000-pixel-tall waterfall image with smooth pan. Overlay masks as SVG in the OSD overlay layer. This single component does more demo work than anything else on the frontend.
Charts	`Recharts` or `visx`	Evidence-provenance stacked bars, reliability diagrams, false-alarm-per-km^2 trend.
State	`Zustand 5`	Minimal. Redux is unnecessary ceremony here.
Data layer	`TanStack Query 5`	Polling for inference job status, optimistic updates on confirm/reject.
Styling	`Tailwind CSS 4` + `shadcn/ui`	Fastest route to a dashboard that does not look like a student project. Dark theme — it reads as an operations console and sonar imagery is grayscale.
Icons	`lucide-react`	Consistent, no licensing questions.
Export	`@turf/turf`, `togpx`	GPX export means the dive team can load targets into a handheld GPS. That detail lands with judges.

4.9 Infrastructure and DevOps

COMPONENT	CHOICE	RATIONALE / NOTES
Containerisation	`Docker` + `Docker Compose v2`	Single `docker compose up`. Services: `db`, `api`, `inference`, `web`. Pre-seed the DB with a demo survey so data appears within five seconds of launch.
Python env	`uv`	10–100× faster than pip. Matters when you rebuild the image fifteen times overnight.
Node env	`pnpm`	Faster installs, strict dependency resolution.
Lint / format	`ruff`, `biome`	Both are single-binary and instant.
Pre-commit	`pre-commit`	Optional but prevents a broken `main` at 3 a.m.

COMPONENT	CHOICE	RATIONALE / NOTES
CI	GitHub Actions	One workflow: lint, unit tests, build images. A green badge on the README is worth the twenty minutes.
Testing	`pytest`, `pytest-asyncio`, `vitest`, `playwright`	Write tests for the geolocation maths and the shadow-geometry module specifically — those are where silent errors hide.
Demo hosting	Local laptop + `cloudflared` tunnel	CRITICAL Never depend on venue wifi for the demo. Run everything locally; the tunnel is only so judges can open it on their phones.
Observability	`structlog` + Prometheus counters	Log inference latency per stage. A latency breakdown chart is a strong answer to "will this run onboard?"

4.10 Edge hardware target

COMPONENT	CHOICE	RATIONALE / NOTES
Reference device	NVIDIA Jetson Orin Nano 8 GB	40 TOPS INT8, 15 W envelope. Realistic for an AUV payload bay. Quote this in the deck even without the hardware in hand.
Alternative	Raspberry Pi 5 + Hailo-8L	13 TOPS, much lower cost. Relevant if the deployment story is a towfish on a fishing vessel rather than an AUV.
OS	JetPack 6.x / Ubuntu 22.04	—
Runtime	TensorRT FP16 → INT8	Report the measured latency and the accuracy delta from quantisation. Honesty here reads as competence.
Power/thermal	Measured, reported	If you have the device, measure watts under sustained inference. Almost nobody does this and it separates you instantly.

05 Model Zoo and Architectural Choices

STAGE	PRIMARY	FALLBACK	WHY THIS OVER THE OBVIOUS ALTERNATIVE
Detection	YOLOv11-s	RT-DETR-R18	YOLOv11-s hits real-time on Jetson and fine-tunes on small datasets. RT-DETR is NMS-free and cleaner architecturally but needs more data to converge — a real risk with 286 AI4Shipwrecks images.
Segmentation	SegFormer-B0	DeepLabV3+/R50	SegFormer's hierarchical attention handles thin elongated structures better than DeepLab's ASPP, and B0 is 3.7M parameters versus 40M+. DeepLabV3+ is kept as fallback precisely because GhostNetZero used it — a directly comparable baseline is worth having.
Artificiality	XGBoost on 24 engineered features	ConvNeXt-Tiny	Trains in seconds, needs no GPU, and produces a SHAP plot. When a judge asks "why did it flag that?", a feature-attribution chart beats a saliency map.
Open-set	Energy-based OOD score	Mahalanobis distance	Free — computed from existing logits, no retraining, no extra head. Threshold tuned on a held-out natural-seabed set.
Embedding	Detector penultimate features → pgvector	DINOv2 features	Reuses compute already spent. Powers similarity search over confirmed targets.
Tracking	Hungarian assignment in geographic space	ByteTrack	ByteTrack assumes video-like motion continuity. Sonar waterfall is not video — successive pings are a moving 1D scan line. Geographic association is both simpler and more correct.

PRETRAINING SEQUENCE

International SSS (AI4Shipwrecks 286 images, SubPipe 10,030 images / 6,335 annotations) → general acoustic representation → synthetic Indian-composite frames → Indian natural-seabed hard negatives → Indian debris fine-tune → held-out unseen Indian region and unseen sonar test. Never randomly split adjacent pings; split by survey mission, geographic region, and sonar configuration.

06 Novel Contributions — The Seven Differentiators

These are the components that create separation from the field. Ordered by leverage per hour invested.

6.1 Shadow-geometry physics verification HIGHEST DEFENSIBILITY

For a target of height `h` observed at slant range `R` from a vehicle at altitude `H`, the acoustic shadow length on the seabed is:

```
# Expected shadow length from sonar geometry
L_shadow = h * R / (H - h)

# Invert: infer implied object height from an observed shadow
h_implied = (L_observed * H) / (R + L_observed)

# Reject if the implied height is physically implausible for the
# hypothesised class, or if no h in [0, H) reproduces the observation.
if not (CLASS_HEIGHT_MIN[cls] <= h_implied <= CLASS_HEIGHT_MAX[cls]):
    candidate.reject(reason="shadow_geometry_inconsistent")
```

Extract the observed shadow by morphological thresholding immediately down-range of the bright return, then fit its extent. Render the predicted versus observed shadow polygon in the UI. This is a hard, explainable, physics-grounded filter that no prior-art model in this space applies, and it visibly kills the rock and coral false positives that would otherwise dominate.

6.2 Synthetic Indian ghost-net generator HIGHEST DATA LEVERAGE

You will not obtain enough labelled Indian SSS data before the finale. Manufacture it. Procedurally generate net meshes, drape them over synthetic seabed relief, compute backscatter with a Lambertian-plus-specular model, cast geometrically correct shadows, apply Rayleigh-distributed speckle, and Poisson-composite onto *real* Indian seabed backgrounds so background statistics remain genuine.

> TIMING
>
> Build this **before** the hackathon. It should be a pre-existing asset you deploy at hour 2, not something you debug at hour 30. A crude version producing 5,000 labelled frames beats a beautiful version that is 70% finished.

6.3 Cleanup-priority scoring — the actual product

```
priority = w1 * calibrated_confidence
         + w2 * entanglement_risk[class]        # gillnet > drum > pipe
         + w3 * reef_proximity_score            # GEBCO + MPA polygons
         + w4 * diver_accessibility(depth)      # peaks near 12–30 m
         + w5 * local_cluster_density           # ST_ClusterDBSCAN
         - w6 * navigation_uncertainty_m        # penalise poor fixes
```

Weights exposed as sliders in the dashboard so an operator can re-rank live. Output is a ranked GeoJSON plus a GPX waypoint file. This converts a computer-vision demo into a dive plan — the difference between a project and a product.

6.4 Evidence-provenance confidence

Every target stores a decomposed evidence vector, displayed as a stacked bar:

EVIDENCE SOURCE	RANGE	CONTRIBUTION
Detector objectness	0–1	Raw YOLO confidence, temperature-calibrated
Shadow-geometry consistency	0 / 0.5 / 1	Fail / indeterminate / pass
Ping persistence	0–1	`min(n_observations / 5, 1.0)`
Artificiality verifier	0–1	XGBoost P(artificial)
Bathymetric plausibility	0–1	Penalty on high-roughness terrain where rock is likely
Open-set novelty	0–1	Inverted energy score; high novelty routes to review

6.5 Human-in-the-loop active learning

Confirm / Reject / Reclassify writes to a labelled queue. The review list is sorted by predictive uncertainty (BALD or simple margin sampling) so the operator labels the most informative frames first. **Demonstrate live that 50 operator clicks measurably improve the model.** If you can show that loop closing on stage, the round is effectively over.

6.6 Frequency-conditioned normalisation (FiLM)

Sonar appearance shifts drastically with operating frequency and range setting. Inject them as conditioning rather than hoping the network infers them:

```
class FiLM(nn.Module):
    def __init__(self, n_cond, n_feat):
        super().__init__()
        self.gamma = nn.Linear(n_cond, n_feat)
        self.beta  = nn.Linear(n_cond, n_feat)

    def forward(self, x, cond):   # cond: [freq_khz, range_m, altitude_m]
        g = self.gamma(cond)[:, :, None, None]
        b = self.beta(cond)[:, :, None, None]
        return x * (1 + g) + b
```

Roughly fifteen lines, inserted after the detector's first two stages. Directly attacks the domain-shift problem that is the acknowledged central challenge.

6.7 Indian hard-negative bank and the operator metric

Curate a dedicated negative set: Gulf of Mannar coral heads, sand ripple fields, rocky outcrops, seagrass beds, sediment mounds, palaeochannel edges. Mine hard negatives iteratively — every false positive the model produces on this bank is added back with a natural label. Then report the headline metric as **false alarms per km² of natural seabed at fixed recall**, alongside conventional mAP. Choosing the operator's metric over the leaderboard metric is itself a contribution and it frames every subsequent question in your favour.

6.8 Sonar-correct augmentation policy

DO NOT USE IMAGENET AUGMENTATIONS

Forbidden: vertical flip (inverts the shadow-return relationship and teaches physically impossible geometry), aggressive rotation (across-track and along-track axes are not interchangeable), hue/saturation jitter (meaningless on acoustic intensity), heavy blur (destroys thin net strands).

Use instead: horizontal flip only (port ↔ starboard is legitimate), along-track stretch and compression (simulates vehicle speed variation), range-dependent gain perturbation (simulates TVG differences), Rayleigh speckle injection at varied intensity, simulated ping dropout, slant-range warp, and shadow-length perturbation consistent with altitude changes.

07 Data Strategy and Acquisition

7.1 Public datasets — pretraining only

SOURCE	CONTENTS	ROLE AND CAVEATS
AI4Shipwrecks	286 high-resolution SSS images, pixel-level shipwreck labels	Acoustic-representation pretraining. Collected in Lake Huron — freshwater, non-Indian seabed. Not evidence of Indian performance.
SubPipe	10,030 SSS images, 6,335 annotations, low/high-frequency pairs, COCO + YOLO labels, navigation and camera metadata	Largest usable volume. Pipeline-inspection domain. The dual-frequency pairs are the best available material for validating FiLM conditioning.
GEBCO_2026 Grid + TID	Global 15 arc-second bathymetry with source-type identifier grid	Regional terrain context only. Download an Indian-water subset as GeoTIFF; the global NetCDF is ~7.0 GB and the TID grid ~3.5 GB.
Sonar Common Target Detection	Assorted public SSS mine/object imagery	Supplementary hard positives for the artificiality verifier.
Marine Debris Archive / MARIDA	Sentinel-2 marine debris labels	Not sonar. Useful only for the regional debris-density prior used in prioritisation.

7.2 Indian institutional outreach

Send these before the hackathon. A single reply — even a partial dataset or a letter of interest — is worth more in the deck than any model improvement.

INSTITUTION	WHAT TO ASK FOR
INCOIS, Hyderabad	Formal Data Requisition Form. Ask explicitly for SSS / MBES / AUV survey data in Indian waters with associated navigation and metadata.
NIOT, Chennai	AUV/ROV sonar and underwater-navigation survey datasets where releasable. Local to you — an in-person visit is plausible and far more effective than email.
NIO, Goa	Marine-geophysical, bathymetric and archaeological survey data.
CMFRI	Fishing-gear occurrence, habitat maps, marine-litter ground truth for taxonomy validation.
SDMRI / TN Fishnet Initiative	Ghost-gear recovery locations and gear taxonomy. Highest-value partner — they have the actual retrieved nets and the recovery coordinates.
Indian Coast Guard / Survey of India	Long shot, but a stated interest letter carries weight.

7.3 Data request specification

Ask for exactly this list, in writing: raw SSS in vendor format (XTF/JSF); processed SSS imagery; per-ping timestamps; sonar frequency and range settings; GNSS/INS/DVL trajectory logs; vehicle altitude and depth; MBES or bathymetry products where available; survey-line identifiers; any existing object annotations; and explicit permission for machine-learning research and competition demonstration.

7.4 Class taxonomy

GROUP	CLASSES
Fishing gear	Ghost net, gillnet, trawl net / cod-end, longline or rope, fishing trap, buoy or float
Marine debris	Plastic container, metal drum, cable, sheet material, miscellaneous debris
Maritime infrastructure	Pipeline, cable, anchor, chain, mooring equipment
Wrecks	Modern wreck, large artificial structure
Natural hard negatives	Rock, coral head, coral rubble, reef edge, sand ripple, sediment mound, seagrass, palaeochannel
Open set	Unknown artificial anomaly — routed to human review, never forced into a known class

08 Database Schema

```
-- PostGIS enabled; geometry stored WGS-84, distance maths in UTM 43N/44N

surveys(
  id, name, region, vessel, sonar_model, frequency_khz,
  range_setting_m, start_ts, end_ts, track geometry(LineString,4326),
  operator_org, licence_note
)

pings(
  id, survey_id, ping_index, ts, position geometry(Point,4326),
  heading_deg, altitude_m, depth_m, slant_range_m,
  channel, nav_uncertainty_m
)

tiles(
  id, survey_id, ping_start, ping_end, channel,
  image_path, width_px, height_px, px_per_m_across, px_per_m_along
)

detections(                              -- per-tile, pre-association
  id, tile_id, bbox, mask_rle, class_pred, detector_score,
  shadow_len_px, shadow_consistent, h_implied_m,
  artificiality_score, ood_energy, embedding vector(256),
  model_sha, created_at
)

targets(                                 -- physical objects, post-association
  id, survey_id, centroid geometry(Point,4326),
  footprint geometry(Polygon,4326), class_final, n_observations,
  first_ping, last_ping, confidence_calibrated,
  evidence jsonb,                        -- provenance breakdown
  depth_m, slope_deg, terrain_roughness, reef_distance_m,
  priority_score, status, reviewed_by, reviewed_at
)

review_events(
  id, target_id, action, old_class, new_class,
  operator, ts, note
)                                        -- feeds active-learning queue

-- Key indexes
CREATE INDEX ON targets USING GIST (centroid);
CREATE INDEX ON pings   USING GIST (position);
CREATE INDEX ON detections USING hnsw (embedding vector_cosine_ops);
```

09 API Surface

METHOD	ENDPOINT	PURPOSE
POST	`/surveys`	Create survey, upload imagery bundle plus metadata CSV
POST	`/surveys/{id}/ingest`	Trigger preprocessing and tiling; returns job id
GET	`/jobs/{id}/stream`	SSE progress stream for the dashboard
POST	`/surveys/{id}/infer`	Run the full pipeline; body selects which stages are enabled

METHOD	ENDPOINT	PURPOSE
GET	`/surveys/{id}/targets`	Paginated targets; filters on class, confidence, priority, bbox
GET	`/targets/{id}`	Full record with evidence breakdown and contributing ping list
GET	`/targets/{id}/similar`	DEMO MOMENT pgvector nearest neighbours
POST	`/targets/{id}/review`	Confirm / reject / reclassify; writes to active-learning queue
GET	`/surveys/{id}/review-queue`	Uncertainty-ranked list for the operator
POST	`/surveys/{id}/prioritise`	Recompute priority with operator-supplied weights
GET	`/surveys/{id}/export`	`?format=geojson|csv|gpx|kml` — dive plan output
GET	`/surveys/{id}/metrics`	False alarms per km^2, recall, calibration curve, per-stage latency
GET	`/tiles/{id}/dzi`	Deep-zoom descriptor for the OpenSeadragon waterfall viewer

10 Repository Structure

```
kadal-netra/
├── docker-compose.yml
├── .env.example
├── README.md                 # architecture diagram + 60-second quickstart
│
├── alai/                     # sonar ingestion + preprocessing
│   ├── readers/              # xtf.py, png_meta.py
│   ├── correct.py            # slant-range, TVG, water column
│   ├── tile.py               # waterfall tiler w/ ping provenance
│   └── tests/test_geo.py     # geolocation maths — test this hard
│
├── synth/                    # synthetic net generator (pre-built)
│   ├── net_mesh.py
│   ├── acoustic.py           # Lambert + specular + Rayleigh speckle
│   ├── compositor.py
│   └── generate.py           # CLI: --n 5000 --out data/synth
│
├── models/
│   ├── detector/             # YOLOv11 + FiLM conditioning
│   ├── valai/                # SegFormer segmentation
│   ├── nizhal/               # shadow physics — pure numpy, no NN
│   ├── verifier/             # XGBoost artificiality + features.py
│   ├── openset/              # energy OOD
│   └── export_onnx.py
│
├── thadam/                   # multi-ping association
├── padai/                    # priority scoring + dive-plan export
├── geo/                      # GEBCO sampling, terrain derivatives
│
├── api/                      # FastAPI
│   ├── main.py  routers/  schemas/  db/  alembic/
│
├── web/                      # Next.js 15
│   ├── app/  components/     # MapView, WaterfallViewer, EvidenceBar
│   └── lib/
│
├── notebooks/                # EDA, ablations, calibration plots
├── data/manifest.json        # SHA-256 per file
└── docs/                     # this report, pitch deck, model card
```

11 Evaluation Protocol

11.1 Splitting discipline

THE SINGLE MOST COMMON FATAL ERROR

Never randomly split adjacent sonar frames. Consecutive pings are near-duplicates; a random split produces inflated scores that collapse in the field. Split by survey mission, then by geographic region, then by sonar configuration. State this explicitly in the deck — judges who know the domain will be listening for it.

SPLIT	TRAIN	TEST	TESTS FOR
Within-region	Indian region A	Indian region A, held-out lines	Basic learning
Cross-region	Gulf of Mannar	Different Indian coast	Geographic robustness
Cross-sensor	Sonar model A	Sonar model B	Domain shift — the real test
Hard-negative	Mixed	Curated coral / rock / palaeochannel bank	False-alarm control
Synthetic-to-real	Synthetic only	Real Indian frames	Validates the generator's worth

11.2 Metrics

CATEGORY	METRICS
Detection	Precision, recall, mAP@50, mAP@50-95, F1
Segmentation	IoU, Dice, boundary-F1 (thin structures need boundary metrics)
Calibration	Expected Calibration Error, reliability diagram, Brier score
Operational	**False alarms per km² at fixed recall** — the headline number
Localisation	Geographic error in metres against reference positions
Temporal	Percentage of true targets confirmed across ≥3 pings
Open-set	AUROC on held-out unseen artificial classes
Latency	Per-stage ms, end-to-end per km of survey, peak memory on target device
Human factors	Reduction in imagery volume requiring manual inspection — the pitch metric

11.3 Ablation table to present

Judges respond to a monotonically improving ablation ladder. Prepare exactly this and show it as one slide:

#	CONFIGURATION	WHAT IT PROVES
1	SSS-only detector, no preprocessing	Naive baseline — what the field submits
2	+ sonar-correct preprocessing and augmentation	Domain knowledge matters
3	+ synthetic data pretraining	The generator earns its place
4	+ FiLM frequency conditioning	Cross-sensor robustness
5	+ segmentation head	Thin-target localisation
6	+ shadow-geometry verifier	Large false-alarm reduction
7	+ artificiality verifier and hard negatives	Further false-alarm reduction
8	+ multi-ping persistence	Detections become physical objects
9	+ bathymetric context	Final filter
10	Full pipeline, edge-quantised	Deployability with honest accuracy delta

12 Edge Deployment Path

Onboard inference matters because AUV acoustic uplink bandwidth is tiny — you cannot stream raw sonar to the surface. Detection must happen in the water so only compact target records are transmitted. Make this argument explicitly; it justifies the entire edge-optimisation track.

STEP	ACTION
Export	PyTorch → ONNX opset 17, dynamic batch axis, simplify with `onnxsim`
Validate	Assert ONNX and PyTorch outputs agree to 1e-4 before proceeding
Optimise	TensorRT FP16 engine; then INT8 with a 500-frame sonar calibration set
Measure	Latency per stage, sustained throughput, peak memory, wall-power draw
Report	Publish the accuracy delta from quantisation. Do not hide it.
Degrade	Design a low-power mode: detector plus shadow verifier only, segmentation deferred to surface post-processing

13 36-Hour Build Plan

PRE-HACKATHON — NON-NEGOTIABLE

Synthetic generator working and 5,000 frames produced. Public datasets downloaded and converted. GEBCO Indian subset clipped. Docker Compose skeleton booting. Institutional emails sent. Baseline detector already trained. Arriving with these means hour 0 starts at ablation #3, not #1.

HOURS	WORKSTREAM	DELIVERABLE AT CHECKPOINT
0–2	Setup, split roles	`docker compose up` green. DB migrated and seeded with a demo survey. Repo scaffolded. Everyone can run everything.
2–6	Preprocessing + tiling	Alai pipeline converting a survey to tiles with ping provenance. Geolocation unit tests passing.
6–12	Detector + FiLM	YOLOv11-s fine-tuned on public + synthetic. ONNX exported. Inference endpoint returning boxes.
12–16	Nizhal shadow verifier	Shadow extraction, height inference, reject logic. Measurable false-alarm drop on the hard-negative bank.
16–20	Segmentation + Thadam	SegFormer masks on candidates. Geographic association producing physical targets in PostGIS.
20–24	Geo context + Padai	GEBCO sampling, terrain derivatives, priority scoring, GeoJSON and GPX export working.
24–30	Frontend	MapLibre target map, OpenSeadragon waterfall with mask overlay, evidence-provenance panel, review buttons wired.
30–33	Ablation + metrics	Run the ladder. Generate every chart for the deck. Record a backup demo video.
33–36	Freeze and rehearse	**CODE FREEZE AT HOUR 33** Rehearse the demo four times. Fix nothing that is not demo-breaking.

13.1 Role split for a six-member team

ROLE	OWNS
Sonar / preprocessing	Alai, tiling, geolocation maths, Nizhal shadow physics
ML — detection	Detector, FiLM, augmentation policy, ONNX export
ML — verification	Segmentation, artificiality verifier, open-set, calibration
Backend / geo	FastAPI, PostGIS, GEBCO sampling, Thadam, Padai, exports
Frontend	Next.js, MapLibre, OpenSeadragon, evidence UI, review loop
Pitch / integration	Deck, ablation charts, demo script, institutional correspondence, backup video. This is a full-time role — do not let it be an afterthought.

13.2 Explicitly out of scope

CUT TensorRT compilation, XTF parsing, MBES fusion, authentication, multi-user support, Celery, Kubernetes, mobile app, real-time AUV telemetry ingest. Each appears in the roadmap slide instead. Naming what you deliberately deferred, and why, reads as engineering judgement rather than omission.

14 Demo Script

Seven minutes. Rehearsed. Nothing typed live except one slider drag.

TIME	BEAT
0:00	One slide: a ghost net photographed in the Gulf of Mannar, and the scale of the problem. Ten seconds, no reading aloud.
0:20	"A survey vessel generates hours of sonar imagery that a human must scroll through frame by frame." Show the raw waterfall in OpenSeadragon, scroll fast. The judges feel the volume problem.
1:00	Run inference. The SSE progress stream ticks through named stages — Alai, detector, Nizhal, Valai, Thadam. Naming the stages makes the architecture legible without a diagram.
1:45	Map populates with ranked targets. Zoom to the top-priority cluster.
2:15	Open one target. Show the evidence-provenance bar. Explain shadow-geometry verification with the predicted-versus-observed overlay. **This is the moment that wins the round.**
3:15	Click a rock the detector initially flagged, show Nizhal rejecting it on implied-height grounds. Demonstrating a correct rejection is more persuasive than another correct detection.
4:00	Similar-target search via pgvector: "find everything that looks like this." Fifteen seconds, high impact.
4:30	Drag the priority weight sliders. Ranking reorders live. Export the GPX. "This file loads into a dive team's handheld GPS."
5:15	Ablation ladder slide. Land on false alarms per km^2 dropping across stages.
6:00	Honesty slide: what we did not claim, what is prior art, what the limitations are, what data we still need. Then the institutional-outreach status.
6:40	Roadmap and close.

> **CONTINGENCY**
>
> Record a full screen-capture of the working demo at hour 32. If anything fails on stage, cut to the video within five seconds without commentary. Practise that transition.

15 Risks, Mitigations and Claim Discipline

RISK	EFFECT	MITIGATION
No Indian SSS labels arrive	Core Indian-domain claim is unsupported	Synthetic generator composited onto real Indian backgrounds; frame the claim as "an Indian-domain adaptation *method* validated on synthetic-to-real transfer", not as a validated Indian model
Class imbalance	Rare debris ignored	Focal loss, hard-negative mining, targeted oversampling, per-class recall reporting

RISK	EFFECT	MITIGATION
Cross-sensor domain shift	Collapse on unseen sonar	FiLM conditioning, aggressive frequency and gain augmentation, explicit cross-sensor test split
Natural false positives	Operator abandons the tool	Nizhal physics filter, artificiality verifier, bathymetric context, multi-ping persistence — four independent filters
Adjacent-frame leakage	Inflated scores, humiliating in Q&A	Survey-level splits, stated explicitly on the metrics slide
Poor navigation metadata	Wrong geolocation	Quality flags, timestamp synchronisation checks, propagate navigation uncertainty into the priority score as a penalty
Model too large for edge	Deployment story collapses	SegFormer-B0, INT8 quantisation, low-power degraded mode
Ultralytics AGPL licence	Productionisation blocker	Disclose in the deck; name RT-DETR as the licence-clean migration path
Demo failure on stage	Round lost	Everything local, hour-32 backup video, code freeze at 33, four rehearsals
Archaeological overclaim	Scientific credibility destroyed	Removed entirely from the pitch. Palaeochannels appear only as a hard-negative class.

16 Post-Hackathon Roadmap

HORIZON	WORK
Weeks 1–4	XTF/JSF native ingestion. Formalise INCOIS and NIOT data requests. Publish the synthetic generator as an open-source package — that alone is a citable contribution.
Months 2–3	Field trial with SDMRI or the TN Fishnet Initiative. Real ground truth from recovered nets with recorded coordinates closes the validation loop.
Months 3–6	Jetson deployment on a towfish. MBES fusion. Multi-survey change detection — "what debris is new since the last survey" is a strong second product.
Months 6–12	Publish the Indian SSS benchmark dataset and the domain-adaptation results. An Indian sonar benchmark does not currently exist; creating one is a durable contribution well beyond the hackathon.

> **THE THROUGH-LINE**
>
> Every technical decision in this document serves one goal: producing fewer, higher-quality, georeferenced, actionable targets rather than maximising an image-level accuracy number. The metric that matters is whether a survey operator can use Kadal Netra to reduce the sonar data they must manually inspect, while keeping high recall on debris that genuinely needs removing. Build for that, pitch that, and measure that.